CSC 3320: System-Level Programming

Review for exam 1

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Note: These slides are updated based on original notes from Raj Sunderraman, Michael Weeks, and Yuan Long

Disclaimer

This review merely highlights some of the important areas that you may focus for exam 1. For complete preparation, please refer to all weekly slides/books/lecture notes from week 1 to week 5.

How much exam 1 will cover?

Introduction

Unix Ch 1

Unix Ch 2

C Basics

- Ch 1
- Ch 2
- Ch 3
- Ch 4

(week 1 to week 5 (lecture 1))

Question Types and Other Essentials

- MCQ: 40 – 45%
  - Conceptual, but with options to choose
- Short-Answer/writing or understanding codes – 50 – 55%. Example:
  - Conceptual, but without options (you will write answers directly)
  - writing a UNIX/Linux command;
  - Interpreting a UNIX/Linux command and their outputs
  - Writing/explaining the semantics of regular expressions
  - Understanding/determining the output of a code snippet
  - Writing (small) programs
  - Explaining a block of code, etc.

Use sftp to upload files

```
(base) Mds-MacBook-Pro:CSC 3320 System Programming mdmahfuzurrahman$ sftp snowball.cs.gsu.edu
mrahman21@snowball.cs.gsu.edu's password:
Connected to snowball.cs.gsu.edu.
sftp> put unix.txt
Uploading unix.txt to /home/mrahman21/unix.txt
unix.txt                                                    100%  188     7.5KB/s   00:00
sftp> put sysprog.txt
Uploading sysprog.txt to /home/mrahman21/sysprog.txt
sysprog.txt                                                 100%   84     3.4KB/s   00:00
sftp>
```

Now, check those uploaded files in SNOWBALL server

```
[mrahman21@gsuad.gsu.edu@snowball ~]$ ls
sysprog.txt  unix.txt
[mrahman21@gsuad.gsu.edu@snowball ~]$
```

For more on sftp, visit this site: https://linuxize.com/post/how-to-use-linux-sftp-command-to-transfer-files/

Downloading files using sftp

- To download a file, use the following command:

```
sftp> get filename
```

- To download a directory, use the following command:

```
sftp> get -r remote_directory
```

Topics

- Benefits of system programming
- Learn the distinction between system programs and system calls.
- Examples of system programs
- Learn about software component and software categories
- Unix architecture
- Process vs. Program distinction
- Unix attributes/advantages
- Unix philosophy

Topics (contd.)

- cat command, pipe mechanism
- head/tail commands
- Script command and terminal recording
- Learn Unix Ch 2 (all parts) very well
  - Attaching few slides from Ch 2 here as examples
  - Study the complete chapter (all slides)

Running Utilities

To run a utility, simply type the name of the utility after the prompt, and press ***Enter*** key

- Some utilities: *date*, *man*, *clear*, *stty*, *passwd*
  - Utility: date [yymmddhhmm[.ss]]
  - Utility: clear
  - Utility: man [-s section] word
    man -k keyword
- To logout, enter CTRL-D (or *exit*)

Special Characters

Backspace

erase: CTRL-H — erase character before cursor

- werase: CTRL-W — erase word before cursor
- kill: CTRL-U — erase entire line
- rprnt: CTRL-R — reprint line
- To test these:
  - type some input like “one two three four” at the terminal prompt
  - move cursor around
  - try the above characters

Special Characters

intr: CTRL-C interrupt running program

- susp: CTRL-Z suspend running program

stop: CTRL-S/CTRL-Q stop printing to screen

- eof: CTRL-D give program end of file
- To test these:
  - Try a command like ***find / (e.g.,*** find . -name "*.txt"***)***
  - For ***eof***, try ***cat > testfile*** then **CTRL-D** on new line

fg command will resume the most recently stopped (suspended) job in the foreground

Some common UNIX utilities

Utilities	Feature
ls	List files
cat ,more, page, head, tail	Show file contents
mkdir, rmdir,cd, pwd	Working with directories
mv, cp, rm	Working with files
wc	Word count
file, groups, chgrp, chmod	About file permission
vi, emacs	Text editor

Pathnames

The hierarchy from a starting directory to a target file or directory.

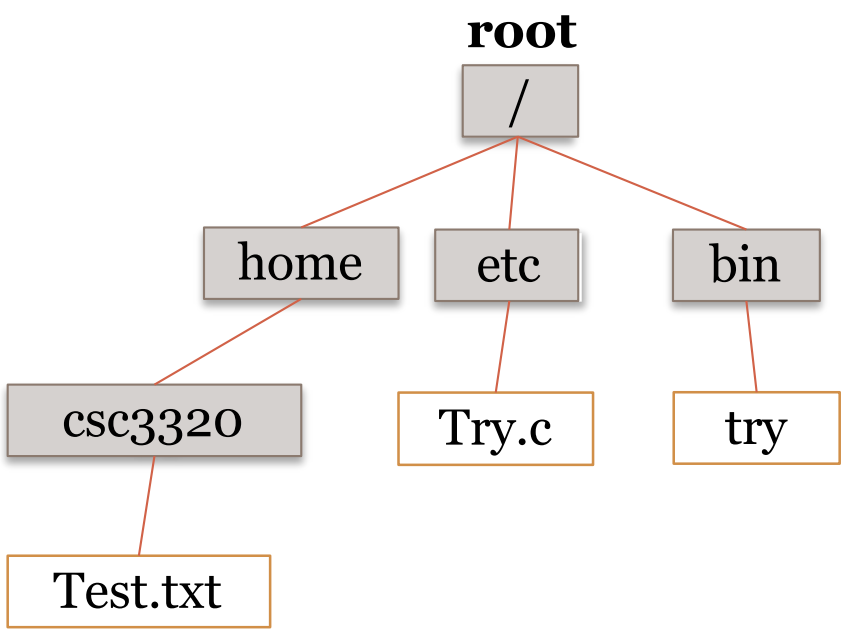
- E.g. /home/myFile , ../myFile, ./myFile

Absolute pathname : relative to the root directory

Relative pathname : relative to its current working directory

/	refers to root directory
.	refers to current directory (example: ./a.out)
..	refers to parent directory (example: ../p2.cpp)

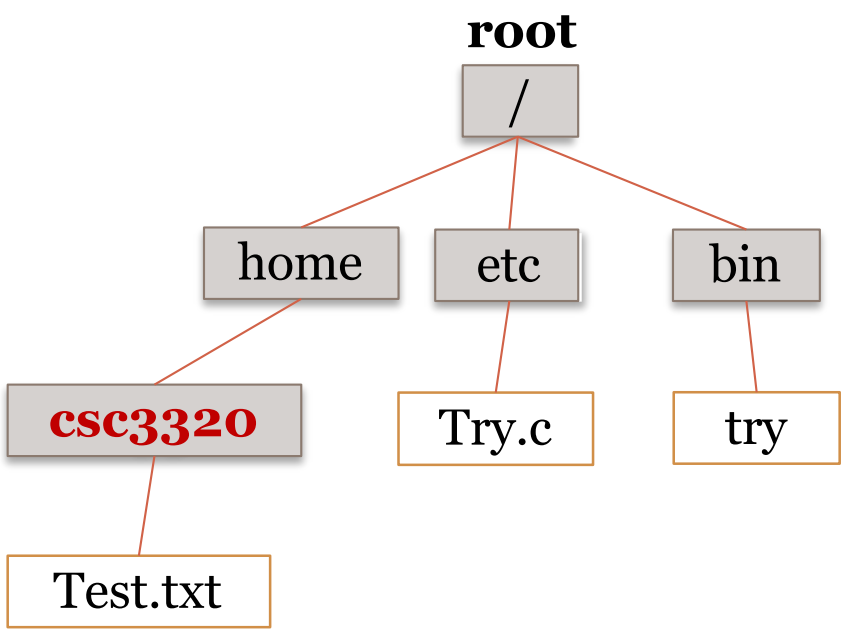
Pathnames



File	Absolute Path
Test.txt	/home/csc3320/Test.txt
Try.c	/etc/Try.c
try	/bin/try

/	refers to root directory
.	refers to current directory (example: ./a.out)
..	refers to parent directory (example: ../p2.cpp)

Pathnames



File	Relative Path
Test.txt	Test.txt or ./Test.txt
Try.c	../../etc/Try.c
try	../../bin/try

Assume csc3320 as the current working directory.

/	refers to root directory
.	refers to current directory (example: ./a.out)
..	refers to parent directory (example: ../p2.cpp)

List Files (*ls*)

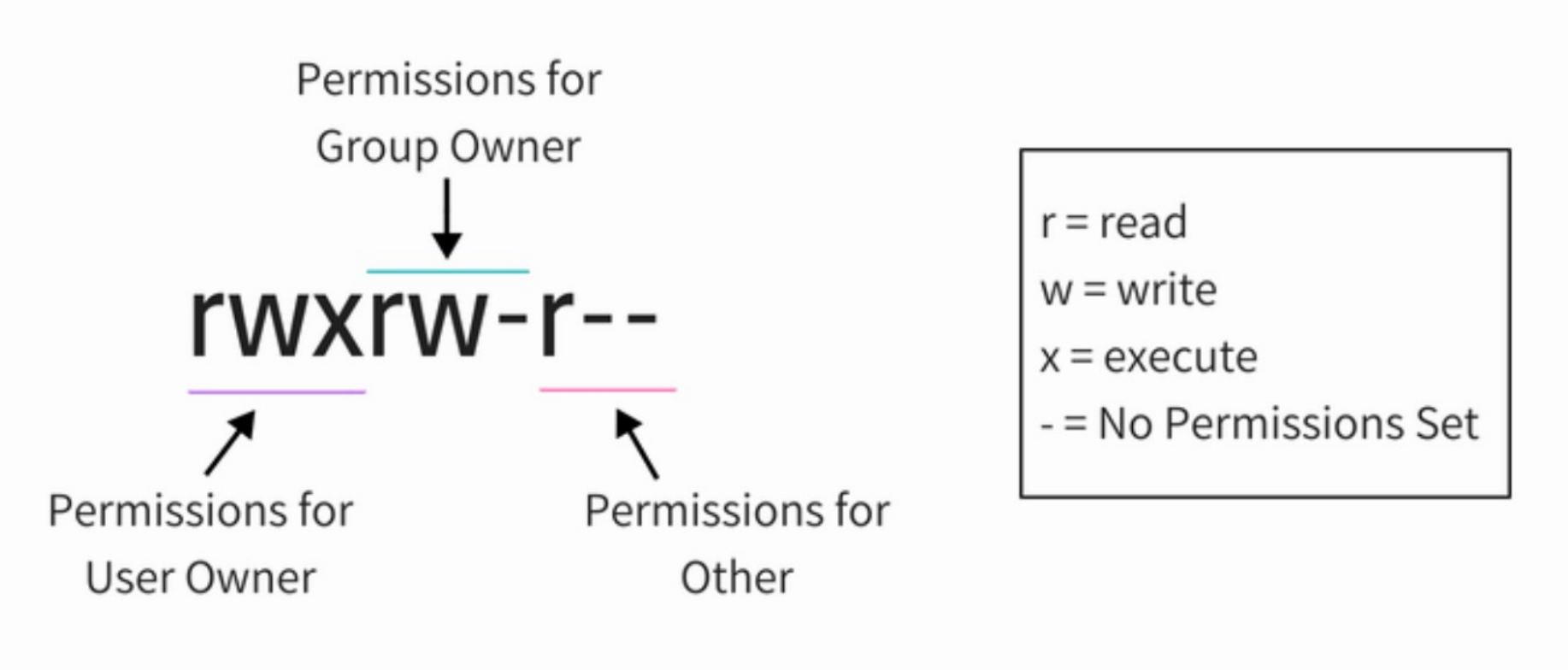
$ ***ls*** *-algFsdR* ***<file-spec-list>***

- options
  - *a*: hidden files
  - *l*: **long listing**
  - *g*: group
  - *F*: put character after file name indicating executable *, link @, directory /, socket =
  - *s*: number of disk blocks
  - *d*: dir details not contents
  - *R*: recursive listing

- The input can be a set of elements, separated by space
- **File-spec** is the pathname for directory or file

How to interpret permissions?

```
(base) Mds-MacBook-Pro:scripts mdmahfuzurrahman$ ls -l test.c
-rw-r--r--  1 mdmahfuzurrahman  staff  131 Sep  1 23:14 test.c
```



Change File Permissions

Another example with octal numbers

```
mweeks@carmaux:~$ ls -l example.txt
-rw-r--r-- 1 mweeks mweeks 51 2007-08-23 16:06
example.txt
mweeks@carmaux:~$ chmod 765 example.txt
mweeks@carmaux:~$ ls -l example.txt
-rwxrw-r-x 1 mweeks mweeks 51 2007-08-23 16:06
example.txt
```

Notice

- 7 = 111 (binary) r “on”, w “on”, x “on”
- 6 = 110 (binary) r “on”, w “on”, x “off”
- 5 = 101 (binary) r “on”, w “off”, x “on”

vi Editor Topics

- Vi editor
  - Movement within vi editor
  - Delete, move, transfer/copy, replace, etc.
  - Quit, write, quit/write, exit/write, etc.
  - Regular expression (Regex)
    - {}, (), [], *, +, ., \ , ^, $

Practices

Write regular expressions for following patterns.

- Phone number with the format ###-###-####
- Date with the format MM/DD/YYYY
- Passwords: strings with at least one upper case letter, one lower case letter, and one digit

Choose all matched strings of the given extended regular expression.

- (1) a(bc)+de
  A) abcde B) ade C) abcbcde D) abc
- (2) [a-z\s]*hello
  A) Othello B) hello C) 2hello D) say hello

C Programming Topics

- Up to Week 5 lecture 1
  - Chapter 1 complete
  - Chapter 2 (general C program structure, compiling, linking, directives, etc.)
    - Example compilation and runs, etc.
  - Chapter 3
    - Printf and scanf function
    - Format specifiers
  - Chapter 4
    - Operators
    - Precedence and associativity
    - Increment/decrement operators

For more details, see sample questions